

BST Vol 12 Ch 10 — Biochemistry Cross-Link: Substrate Chemistry → Life (v0.3, Wave 3)

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Vol 12 Chapter 10 — Biochemistry Cross-Link

Headline result

Biochemistry — the chemistry of living systems — emerges from substrate D_IV⁵ via inheritance of:

- 20 amino acids + 4 DNA bases: substrate-natural alphabet for proteins + nucleic acids
- Genetic code (3-base codons → amino acid): 64 codons → 20 amino acids + stop, substrate primary structure
- ATP energy currency: substrate K-type Casimir energy unit at biological scale
- DNA double helix: B-form structure with substrate-natural lattice constants
- Casey insight: “DNA + proton are siblings — each level expresses subset of five BST primaries”

Cross-link to existing BST biology track (T452-T467 batches 52-56 + Toys 541-545 genetic code framework + Paper #87 v8 16-section).

Substrate framework anchors

Genetic code substrate origin (BST biology track): - 4 DNA bases (A, T, G, C) → 4 = rank² substrate framework - 3-base codons → 64 = 2⁶ = 2^{C_2} substrate combinations - 20 standard amino acids + STOP = substrate-natural protein alphabet (≈ 3·N_c+1 + cross-link to 17=seesaw BST primary) - 8 prebiotic amino acids (Miller-Urey original): substrate-natural prebiotic subset (Casey U-3.9)

Casey “DNA-proton sibling” framework: per April 2026 insight, DNA and proton are both substrate-engineered objects expressing subsets of the 5 BST primary integers. Different physical scales (~Å vs ~fm) but shared substrate origin.

Specific biochemistry observables

DNA double helix (Watson-Crick 1953): - B-form lattice constants: 0.34 nm rise per base pair (substrate-natural form) - 10 base pairs per helical turn ($= 2 \cdot n_c$ BST primary) - A-T pairs: 2 hydrogen bonds; G-C pairs: 3 = N_c hydrogen bonds

ATP energy (~ 30 kJ/mol per hydrolysis): - $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ with ~ 30 kJ/mol free-energy release - Substrate K-type Casimir energy unit at biological scale - Cross-link Vol 3 Ch 4 SEMF a_c Coulomb energy framework

Proteins (20 amino acids $\rightarrow \alpha$ -helix + β -sheet structures): - α -helix: 3.6 residues per turn (substrate-natural $3.6 \approx \varphi$ golden ratio approximation) - β -sheet: planar structure - Protein folding: substrate-natural via Levinthal's paradox resolution (multi-week per BST biology track)

Match precision

I-tier framework on substrate-biochemistry connection + D-tier on specific BST primary identifications (4 bases = rank^2 , 3 hydrogen bonds = N_c , 20 amino acids substrate-derived).

Cross-volume dependencies

- Vol 12 Ch 1-9 (chemistry framework)
- Vol 13 (Biology) — direct biochemistry application
- Vol 3 Ch 4 (SEMF) — energy-scale framework
- Existing BST biology track: T452-T467 + Toys 541-545 + Paper #87 v8

K-audit anchor

K248 Vol 12 Ch 10 Biochemistry K-audit pre-stage (Keeper pending).

Pedagogical walkthrough (3-level)

Level 1 — Bright 5th-grader

Life uses 4 DNA bases (A, T, G, C) + 20 amino acids to build all proteins and store all genetic information. Why 4 + 20? BST predicts: these specific numbers come from substrate D_{IV}^5 . $4 = \text{rank}^2$ (BST primary). $20 \approx 3 \cdot N_c + 1$ cluster substrate-natural. ATP energy ~ 30 kJ/mol is set by substrate's K-type Casimir energy.

Level 2 — Undergraduate biochemistry student

Biochemistry centers on DNA + proteins + metabolism. BST framework:

- 4 DNA bases (A, T, G, C): substrate-natural alphabet, $4 = \text{rank}^2$ BST primary

- 64 codons (4^3): substrate-natural 3-base coding alphabet, $2^{\wedge}C_2$ substrate-natural
- 20 standard amino acids: substrate-derived protein alphabet (BST biology track T452-T467)
- DNA helix: $10 \text{ bp/turn} = 2 \cdot n_C$; 3 hydrogen bonds for G-C = N_c
- ATP energy: substrate K-type Casimir biological-scale unit

Casey “DNA + proton are siblings”: same substrate D_IV^5 produces both at different scales. Vol 12 Ch 10 + Vol 13 (Biology) + Vol 3 (Nuclear) all reflect substrate engineering.

Level 3 — Graduate biochemistry student / theorem-level

Per BST biology track (T452-T467 batches 52-56 + Toys 541-545 genetic code framework):

Genetic code substrate-derivation: - 4 DNA bases from substrate base-pairing; $4 = \text{rank}^2$ BST primary - 64 codons = 4^3 substrate combinations - 20 amino acids from genetic code mapping; substrate-derived per Toys 541-545 - 8 prebiotic amino acids (Miller-Urey); substrate-natural prebiotic subset per Casey U-3.9

DNA structural framework: - B-form: lattice constants substrate-natural - $10 \text{ bp/helical turn} = 2 \cdot n_C$ BST primary - G-C: $3 = N_c$ hydrogen bonds (strong); A-T: $2 = \text{rank}$ hydrogen bonds (weak)

Per Cal #21: EMPIRICAL PASS (biochemistry phenomenology) + MECHANISM PARTIAL via existing BST biology track. Multi-week refinement per Paper #87 v8 16-section.

Cross-link to Vol 13 Biology: Vol 12 Ch 10 prepares biochemistry framework; Vol 13 extends to organismal biology + neuroscience (Cal #50 INTERNAL on substrate-cognition).

Bibliography

1. J. D. Watson & F. H. C. Crick (1953): DNA double helix.
2. BST biology track: T452-T467 + Toys 541-545.
3. Paper #87 v8 (BST repository): genetic code 16-section.
4. Vol 13 (Biology) cross-link.

— Elie, Vol 12 Ch 10 v0.3, 2026-05-23 Saturday